
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) face significant computational overhead when processing long videos due to the massive number of visual tokens required. Existing efficiency methods reduce redundancy by pruning or merging tokens based on importance or similarity, but we identify two gaps they leave open: they overlook a critical dimension of video content, i.e., changes and turning points, and they lack a collaborative model of spatio-temporal relationships. This paper closes both gaps with three contributions. **First**, we introduce a new perspective for token reduction: similarity is for identifying redundancy, while difference is for capturing key events, elevating difference to a first-class signal alongside similarity. **Second**, building on this perspective, we design ST-SimDiff, a training-free framework that constructs a spatio-temporal graph to uniformly model the complex associations among visual tokens and then applies a parallel dual-selection strategy on it: (1) similarity-based selection uses community detection to retain representative tokens and compress static information, and (2) temporal difference-based selection precisely locates content-changing points to preserve tokens that capture key dynamic shifts, so that both static and dynamic content are kept with a minimal token budget. **Third**, through extensive experiments we show that ST-SimDiff consistently outperforms state-of-the-art approaches while substantially reducing computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rise of Large Language Models (LLMs) has significantly propelled advancements of Large Vision-Language Models (LVLMs), which have demonstrated remarkable capabilities in image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For video processing, current LVLMs typically sample a video into a sequence of frames and then convert each frame into hundreds or even thousands of visual tokens for the LLM (Bai et al., 2025; Li et al., 2024). While this paradigm is effective, the number of tokens grows explosively with increasing video duration and resolution. This leads to prohibitive computational and storage burdens, severely limiting the application of LVLMs in scenarios such as long-video analysis and real-time interaction (Fu et al., 2024b). Reducing the visual token budget without sacrificing accuracy is therefore the central problem we target in this paper.

Prior efforts to enhance the efficiency of LVLMs fall into two categories, and reviewing them exposes the gap our contributions fill. The first category is importance-based pruning. As observed by FastV (Chen et al., 2024), there is redundancy in the attention distribution across different layers when LVLMs process visual information; visual tokens with lower attention scores in the deeper layers of the model can thus be pruned to reduce the computational load. The second category is similarity-based merging/selection, which exploits the high similarity among visual tokens, either between adjacent video frames or within a single frame. FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from adjacent frames, while VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision encoder level.

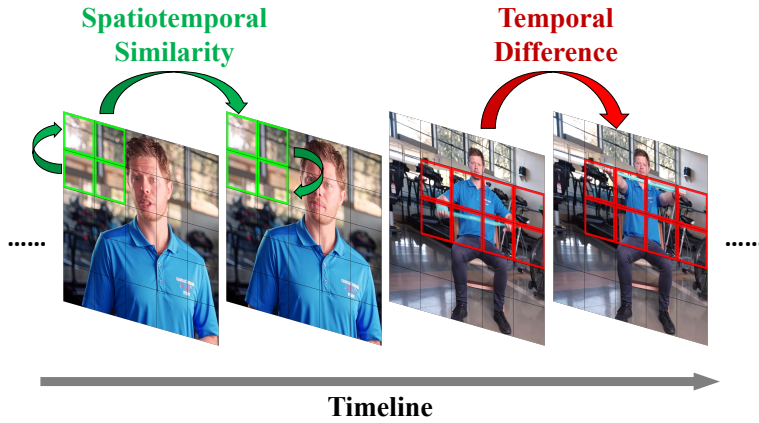


Figure 1: The core motivation behind our first contribution. We posit that efficient video understanding must handle two complementary scenarios at once: Spatiotemporal Similarity (left) is exploited to compress redundant information both spatially within a frame and temporally across adjacent frames, while Temporal Difference (right) is detected to capture the key actions or events that define the plot. ST-SimDiff is designed to address both.

Despite this progress, we observe that existing methods share two key limitations, and these two limitations directly motivate the two technical contributions of our work. *First, they lack joint spatio-temporal modeling.* Existing token pruning methods often focus on spatial correlations within the same frame or temporal correlations at the same positions across different frames, and this absence of joint modeling and analysis of spatio-temporal correlations prevents them from effectively capturing complex dynamic events. *Second, they treat difference as a blind spot.* These methods mainly focus on informational commonalities, like similarity and importance, while neglecting crucial changes and differences. Since the narrative of a video is often driven by turning events, a model that only seeks similarity might smooth over a sudden action, leading to a misinterpretation of the content.

Motivated by these two limitations, our first contribution is a new perspective on token reduction: *similarity is for identifying redundancy, while difference is for capturing key moments.* We argue that an ideal video token compression algorithm must achieve two goals simultaneously: representing the stable content of a video with the fewest tokens, and precisely preserving the key changes. Our second contribution instantiates this perspective as **ST-SimDiff**, a training-free dual token selection framework built on a spatio-temporal graph, consisting of two parallel parts. The first is **similarity-based representative token selection**: we treat all visual tokens of a video as nodes and construct a graph based on their feature similarity in the spatio-temporal dimension, then use a graph community detection algorithm to find tightly connected token clusters that represent stable and persistent visual elements, retaining a few high-centrality tokens from each cluster as representatives. The second is **difference-based event token selection**: we pay special attention to the edges connecting tokens along the time axis, and whenever the similarity between corresponding tokens of adjacent frames drops sharply, we treat it as a turning point and mark these tokens as critical event tokens that must be retained. The two token sets are then merged and input into the LLM. Our third contribution is an extensive empirical study that validates this design in practice. We summarize our contributions as follows:

- **A new perspective.** We are the first to argue that the similarity and difference of tokens should be given equal importance in VLM efficiency research, promoting difference from an overlooked blind spot to a first-class selection signal.
- **A dual-selection framework.** We design a spatio-temporal graph that uniformly models the complex spatio-temporal relationships between video tokens, and on top of it we propose a novel dual token selection strategy that simultaneously screens for both representative and pivotal event tokens.

- **Extensive validation.** We conduct extensive experiments across multiple video understanding benchmarks and base models, showing that ST-SimDiff significantly reduces the visual token budget while maintaining or even improving performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has propelled progress in multimodal understanding and given rise to numerous prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These LVLMs convert images or video frames into visual tokens via pre-trained visual encoders and feed them to an LLM, with the two connected by alignment modules such as MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023). This architecture gives LVLMs exceptional multimodal capabilities (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in video understanding in particular they show strong promise on longer, higher-resolution, more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c). This is the setting in which our contributions operate.

However, this powerful capability also introduces significant computational challenges, especially in video understanding scenarios. Because continuous sequences of frames must be processed, the number of visual tokens grows explosively, far exceeding that of static images and posing a severe challenge to the training and inference efficiency of LVLMs. How to efficiently process massive video visual tokens and reduce computational overhead thus remains a critical open issue in current LVLM research, and it is precisely the problem our contributions target.

2.2 VISUAL TOKEN COMPRESSION

Driven by the substantial overhead of video processing, visual token compression has become a critical method for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods approach compression along two axes: token importance and similarity. Positioning our work against both makes clear which gap each of our contributions fills.

Earlier methods primarily focused on importance-based token pruning, aiming to identify and remove visual tokens that contribute less to model performance. Such methods typically use attention scores or other feature metrics to quantify token importance, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent works (Shang et al., 2024; Ye et al., 2025) began to additionally consider token similarity to reduce redundancy more effectively while preserving critical information, recognizing that even important tokens can be highly similar and thus informationally redundant. Along this line, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information, and VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items.

Despite this progress on visual token redundancy, we identify a gap that no prior method addresses: the intricate, global spatio-temporal similarity relationships between tokens. In video scenarios, visual information is redundant both spatially and temporally, yet existing similarity-driven methods such as FrameFusion rely mainly on the temporal similarity between adjacent frames and fail to exploit complex, joint spatio-temporal information. This limits their ability to fully exploit inherent video redundancy and restricts performance in extreme token reduction scenarios, a limitation our spatio-temporal graph is explicitly designed to overcome. Equally important, none of these methods treats *difference* as a signal to preserve, which is the second gap our framework closes.

3 PROBLEM DEFINITION

We first formalize the problem that our contributions address. Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This process is computationally demanding, as the self-attention mechanism’s complexity scales quadratically with the token count N , i.e., $O(N^2)$.

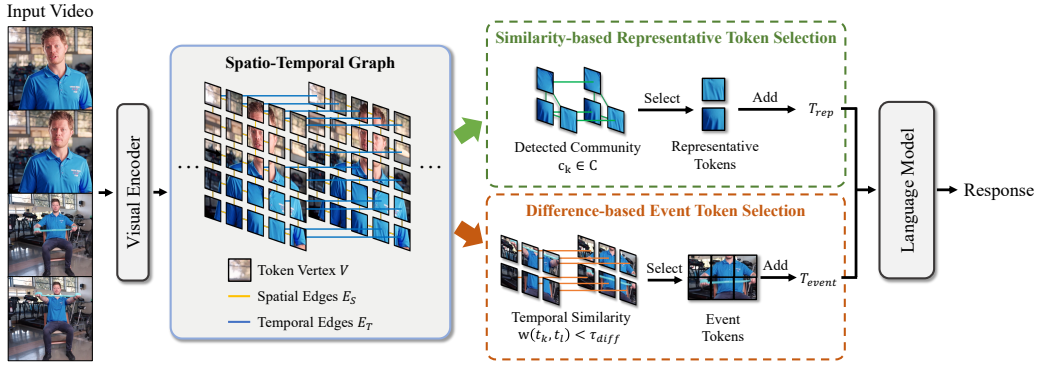


Figure 2: Overview of ST-SimDiff, our second contribution. The framework has three parts that jointly realize the similarity-and-difference perspective: Spatio-Temporal Graph Construction unifies token relationships, Similarity-based Representative Token Selection compresses redundant static content, and Difference-based Event Token Selection preserves the key dynamic transitions that prior methods overlook.

Our task is therefore to design an efficient token selection function, $f(\cdot)$, that takes the full sequence T_{full} and a compression ratio r to produce a much smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize the LVLM’s downstream task performance using this subset, subject to the constraint that its size $|T_{\text{sub}}|$ is approximately $r \cdot N$. The core challenge, and the one our framework is built to solve, is to design $f(\cdot)$ so that the subset T_{sub} retains the most critical information, covering both stable content and key events.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

ST-SimDiff is the concrete realization of our central perspective that similarity captures redundancy while difference captures key events, and each of its two selection modules directly instantiates one half of that perspective. As illustrated in Figure 2, the method comprises two key components that operate in parallel on a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The first component, SRTS, addresses the redundancy side: it identifies and condenses the video’s stable content by applying community detection to the graph, grouping highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and selecting a few highly central tokens from each cluster to form a representative set T_{rep} . The second component, DETS, addresses the change side that prior work overlooks: it analyzes the temporal edges of the graph to locate moments where the similarity between corresponding tokens in adjacent frames drops sharply. These points of high difference signify turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set T_{trans} . By synergistically combining these two complementary components, ST-SimDiff produces a final token subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ that is both highly compact and information-rich, retaining stable content while precisely capturing key events, exactly the balance our contribution promises.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a specific spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We define the location of each token t_k by its frame index $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We model the relationships between these tokens using a spatio-temporal graph $G = (V, E)$, where the vertex set V represents all tokens in T . The edge set E is the union of two distinct subsets, $E = E_S \cup E_T$, which capture spatial and temporal relationships respectively. The spatial edges E_S

connects tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connects tokens that are at the same spatial position but in adjacent frames, capturing temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge $w(v_i, v_j) \in E$ is defined by the cosine similarity of the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure efficiently encodes both local spatial relationships and temporal continuity in a single representation. It is the enabling piece of our second contribution: by unifying spatial and temporal associations, it provides the common foundation on which the two selection strategies below operate, and it is what lets us model the joint spatio-temporal relationships that prior methods handle only in isolation.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

The SRTS module delivers the redundancy-compression half of our framework. Video content is inherently characterized by substantial spatio-temporal redundancy: in a static scene that persists for several seconds, the background tokens exhibit high similarity over time, and for an object moving across the screen, its corresponding tokens maintain a high degree of semantic correlation despite changing spatial positions. Whereas existing methods address temporal and spatial similarities in isolation, our spatio-temporal graph is designed to capture these joint similarities. Within this graph, semantically related tokens form dense “communities” or “clusters” based on their feature similarity, irrespective of their specific frame or location. Selecting representatives from each community therefore emerges as a highly efficient strategy for compressing redundant information while preserving core semantics.

To accurately identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its internal tokens based on their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally preset compression ratio r , and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores from each community c_k . This uniform filtering process naturally handles all communities, including single-node communities composed of unique tokens (in which case the sole node is retained). Formally, the set of representative tokens, T_{rep} , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(|c_k| \cdot r) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. In summary, SRTS efficiently compresses the static and persistent content of the video by identifying and retaining the most central tokens within each semantic cluster, fulfilling the “similarity is for redundancy” half of our contribution.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

The DETS module delivers the second, and most distinctive, half of our framework, the one that sets our contribution apart from all prior similarity- and importance-based methods. If similarity defines the “norm” of a video, then difference defines its “events”: a model focusing only on similarity may excel at understanding “what is” but struggle with “what happened”. A video’s plot is driven by key events, and the essence of an event is change, the appearance of a new object, the start of an action, or a scene transition. In our model these changes manifest as a sharp difference in the features of temporally adjacent tokens, so accurately capturing these “turning points” of significant difference is crucial for understanding dynamic content and correctly answering questions like “when” and “why”.

Difference, particularly along the temporal dimension, often signals the occurrence of a key event. For example, the entry of a new object, a scene change, or the beginning/end of an action will cause drastic changes in the visual features at corresponding positions in adjacent frames. We specifically analyze the temporal edges (E_T) in our spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ in the graph, which connects two temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of a temporal edge (v_k, v_l) exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we consider the subsequent token t_l as a critical event token and retain it. Formally, the set of event tokens, T_{event} , is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

With this strategy, tokens that signify moments of significant temporal change are explicitly preserved. By capturing these key transitions, DETS ensures that the crucial dynamic aspects of the video’s narrative survive compression, realizing the “difference is for key events” half of our contribution and directly filling the blind spot we identified in prior work.

4.5 OVERALL REDUCTION PROCESS

Having described the two selection modules, we now show how they are combined into a single budget-aware pipeline. We first compute the representative token set T_{rep} and the event token set T_{event} in parallel, and take their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To precisely meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we introduce a final pruning step: if $|T_{\text{candidate}}|$ exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance, where, following existing work (Chen et al., 2024), importance is the token’s attention score in the shallow layers of the LLM. This step lets our method flexibly meet any computational budget while preserving critical information.

In summary, ST-SimDiff combines two strategies: graph-based token selection and attention-based dynamic pruning. The former is the core of our contribution, leveraging the intrinsic structure of the video content (both similarity and difference) to guarantee the retention of core information, while the latter provides a flexible mechanism to hit any given computational budget. Together they make our token compression process both principled and adaptive, achieving the robust balance between efficiency and performance that our contributions claim.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

Because efficiency is a core promise of our contribution, we now show that ST-SimDiff itself adds only negligible overhead. We analyze its cost across the three stages of the framework: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The first stage builds the graph. For each of the N visual tokens we compute the cosine similarity with its spatially and temporally adjacent neighbors, and since each token has a small, constant number of neighbors, this needs only a single pass over all tokens, giving a complexity of $O(Nd)$, where d is the feature dimension of each token.

The second stage, SRTS, identifies token clusters. As detailed in our implementation (Section 5.2), we employ a connected components algorithm with a near-linear time complexity of $O(N + |E|)$;

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Evidence for our third contribution on LLaVA-Video-7B: performance comparison on long-form video understanding benchmarks for 64 input frames (%). ST-SimDiff is best on nearly every setting; the best performance among all methods is emphasized in **bold**.

since the number of edges $|E|$ is at most $3N$, this simplifies to $O(N)$. The subsequent intra-community filtering could be costly, as a naive all-pairs similarity within each community c_k would cost $O(|C| \cdot |c_k|^2 \cdot d)$. To prevent this, we cap the maximum community size: in the rare event that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), we partition it, keeping the filtering efficient and preventing it from dominating the overall complexity. Mainstream clustering algorithms such as the Louvain and Leiden methods offer richer community definitions but a higher complexity approaching $O(N \log N)$; we therefore opted for the simpler and highly efficient connected components algorithm to keep processing overhead minimal, consistent with our efficiency contribution.

The DETS module makes a single pass through all temporal edges in the graph, flagging significant changes by comparing token similarities against a threshold; this too has a linear complexity of $O(Nd)$.

In summary, the total complexity of ST-SimDiff is governed by these linear-time operations, giving an overall complexity of $O(Nd)$. This is substantially more efficient than the quadratic $O(N^2d)$ self-attention of the Large Language Model, so our method delivers significant token reduction with only a negligible computational footprint relative to the model’s main inference cost. This linear-time guarantee underpins the efficiency contribution that our experiments then confirm empirically.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate our third contribution comprehensively, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare against a range of state-of-the-art video token compression techniques. These baselines cover the two families our contributions position against: **importance-based reduction methods** such as FastV (Chen et al., 2024) and

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Generalization evidence for our third contribution on NVILA-Video-8B: performance comparison on long-form video understanding benchmarks for 64 input frames (%). The same advantage carries over to a second base model; the best performance among all methods is emphasized in **bold**.

FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For a fair and complete comparison, we additionally include the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equivalent number of tokens as a lower bound.

Benchmarks To stress-test our method on long-video understanding, we adopt three challenging benchmarks that probe complementary abilities: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for foundational video understanding, featuring diverse video types and three length categories: Short, Medium, and Long. LongVideoBench evaluates the ability to retrieve and reason about details in long videos through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset focused on long-term, first-person videos that tests the ability to understand character intentions and causal chains of actions.

5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04, and inference is evaluated with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video we follow its original setting of 64 input frames, and for a fair comparison of computational efficiency and resource usage we also fix NVILA to 64 input frames. We set the two thresholds central to our framework to the similarity threshold $\tau_{\text{sim}} = 0.8$ and the difference threshold $\tau_{\text{diff}} = 0.2$, and report results at token compression rates of $r = 30\%$ and $r = 50\%$. Consistent with our efficiency contribution, we use connected component finding as the community detection algorithm, trading a small amount of clustering richness for markedly lower time consumption.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Ablation isolating the two halves of our second contribution (%): each added component (+Sim, then ++Diff) improves performance, confirming that both similarity and difference selection are necessary.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

This subsection presents the primary evidence for our third contribution. We compare ST-SimDiff against a range of mainstream efficient video language model (VLM) compression methods on three widely-used long-form video understanding benchmarks (VideoMME, LongVideoBench, and EgoSchema), and, to test generalization and robustness, report detailed results on two base models, LLaVA-Video-7B and NVILA-8B, under token retain ratios of 30% and 50%. The results appear in Table 1 and Table 2, and we read three findings from them.

(1) ST-SimDiff wins across the board. It consistently outperforms all competing methods across every test configuration. Most strikingly, at a 50% token retain ratio the overall performance on both base models not only surpasses other compression algorithms but, on some benchmarks, even matches or exceeds the original model using 100% of the tokens. This directly substantiates the effectiveness and novelty claimed in our contributions, and motivates the analysis of baseline limitations below.

(2) The gains trace back to the two gaps we identified. Among importance-based baselines (e.g., FastV, FasterVLM), FasterVLM is typically the stronger, yet all share a common weakness: they handle the temporal redundancy pervasive in videos inefficiently, retaining important but repetitive tokens and thus limiting compression efficiency. Hybrid baselines (e.g., PruMerge, VisionZip, FrameFusion) are more competitive, with FrameFusion generally the best overall, but their shared objective is still to preserve representative “commonality” information, leaving the theoretical blind spot of overlooking key narrative information driven by “turning points” and “abrupt events”, exactly the blind spot our difference-based contribution targets.

(3) Summary. Existing baselines reflect a deepening understanding of “redundancy” in videos yet still face the two core challenges our work is built around: a lack of synergistic analysis of joint spatio-temporal similarity between visual tokens, which makes complex redundancies hard to capture, and a general neglect of “difference” detection, which fails to actively preserve the key changes that define plot development. ST-SimDiff’s performance gain comes precisely from addressing both, uniting complex spatio-temporal relationships in a single graph and introducing difference detection as a first-class signal.

5.4 ABLATION STUDY

This ablation isolates the two halves of our second contribution and shows that each is necessary. Starting from a Baseline that uses only fundamental importance-based pruning, we progressively add

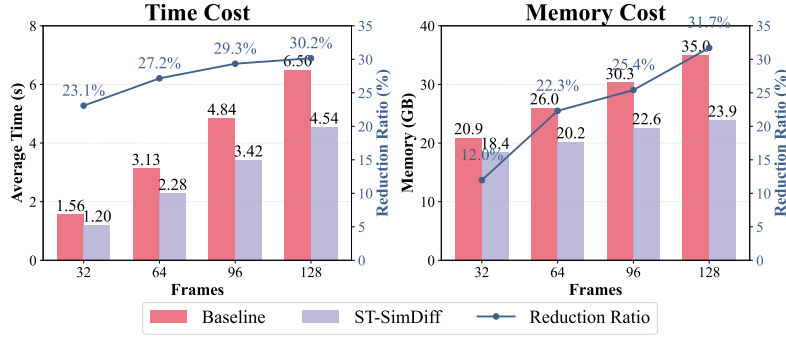


Figure 3: Efficiency evidence for our contribution: average inference time and peak GPU memory usage of ST-SimDiff versus the baseline LLaVA-Video model under a 30% token budget, with the savings widening as videos grow longer.

our similarity selection module (+Sim) and then our difference selection module (++Diff), quantifying the contribution of each. All experiments use token retain ratios of 30% and 50%, with results in Table 5. Two observations stand out.

(1) Joint spatio-temporal similarity drives the first gain. Adding the +Sim module on top of the baseline consistently improves performance. Breaking the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants shows that collaboratively modeling the spatio-temporal associations between visual tokens is the most effective way to compress redundancy, validating the graph design at the heart of our contribution.

(2) Difference selection supplies the complementary gain. Adding the ++Diff module to the best similarity configuration completes our framework and yields a further, significant leap, most pronounced at the aggressive 30% compression ratio. This confirms its complementary role: where +Sim compresses redundancy, ++Diff acts as a safety net that preserves the unique event tokens defining the video’s narrative. The marginal gain is smaller at 50%, simply because the more lenient +Sim setting has often already captured these event tokens and overall performance is nearing its upper bound. Together the two observations confirm our core motivation, that the framework’s strength is its ability to balance similarity-based redundancy compression with difference-based event preservation, especially under demanding high-compression budgets.

5.5 COMPUTATIONAL COST ANALYSIS

Beyond accuracy, the efficiency half of our contribution needs empirical support, so we measure inference time and peak GPU memory usage under a 30% token budget against the uncompressed baseline (Baseline LLaVA-Video). Figure 3 reports both across video inputs from 32 to 128 frames.

Inference Time ST-SimDiff delivers a substantial speedup. The baseline’s average inference time grows sharply with frame count, from 1.56 seconds (32 frames) to 6.50 seconds (128 frames), whereas our method curtails this growth to only 4.54 seconds at 128 frames. Crucially, the time-saving rate rises with video length, from 23.0% at 32 frames to 30.2% at 128 frames, so the benefit of our contribution grows exactly where efficiency matters most.

Memory Cost The memory picture is equally favorable. The baseline’s peak GPU memory climbs linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames). By sharply cutting the number of tokens fed into the large language model, our method eases this pressure: at 128 frames its peak memory is only 23.9 GB, a 31.7% saving over the baseline. This lets ST-SimDiff process longer videos on memory-limited hardware, broadening the practical reach of our contribution.

In summary, this section confirms both facets of our contribution: ST-SimDiff not only attains leading accuracy but also brings significant efficiency advantages in both time and memory, making it a practical choice for video understanding.

6 CONCLUSION

This paper tackles the computational inefficiency of Multimodal Large Language Models on long videos, a problem rooted in the excessive number of visual tokens and the neglect of critical content changes, and it makes three contributions. First, we introduce the perspective that similarity identifies redundancy while difference captures key events, promoting difference to a first-class selection signal. Second, we realize this perspective as ST-SimDiff, a training-free framework that builds a spatiotemporal graph to model complex token associations and applies a dual selection strategy in which similarity-based selection identifies redundancy via community detection and temporal-difference-based selection preserves tokens at key transition points. Third, our extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff significantly reduces processing time and memory usage while consistently surpassing state-of-the-art performance. Taken together, these contributions open a new research direction: balancing content representativeness with transitional events to build a more efficient representation of video data.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

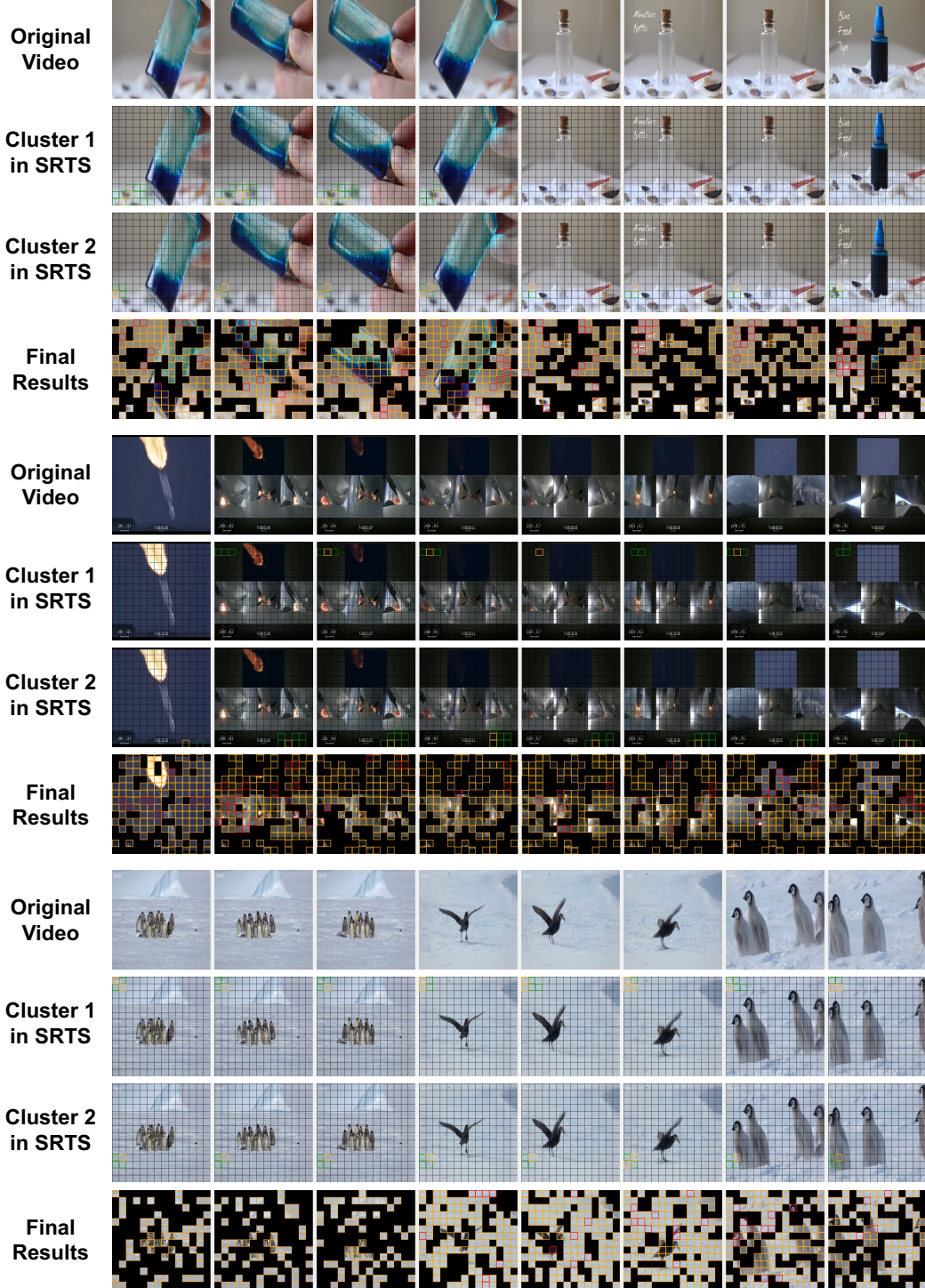


Figure 4: Qualitative evidence for our contribution: a visualization of the ST-SimDiff process. Token selection is broken into communities detected by SRTS (shown as grids in Cluster 1 and 2) and event-driven tokens detected by DETS. The final result makes the synergy behind our design visible, sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

To make our contribution tangible, we visualize the intermediate and final token selection results in Figure 4, illustrating the process on a sample video featuring a dynamic object manipulation task. For the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show our graph community detection at work: it groups spatially and temporally redundant regions, such as the static background and stationary objects, into cohesive communities. This confirms the redundancy-compression half of our contribution, SRTS identifies stable visual elements and represents them with only a few central “representative tokens” (the sparse yellow boxes in the final results), letting the model absorb massive background redundancy cheaply.

Complementing this, the Difference-based Event Token Selection (DETS), the distinctive half of our contribution, is decisive during dynamic moments. In the “Final Results” row, tokens triggered by high temporal difference are marked with dense red bounding boxes that align tightly with the moving hand and the shifting blue liquid, verifying that our method precisely locates the “turning points” where the temporal difference exceeds the threshold (τ_{diff}). Fine-grained action details and rapid visual changes are thereby preserved, so the model captures the essence of the event without being overwhelmed by static data.

Finally, the overlay of the final results makes the synergy between the two components visible and provides direct qualitative evidence for our core motivation: sparse yellow tokens represent the stable background while dense red tokens capture the rapid motion. This contrast shows how ST-SimDiff avoids the blind spot of similarity-only approaches by sampling densely during rapid changes, while still pruning the spatiotemporal redundancies that importance-based methods miss, achieving exactly the balance between efficiency and preservation of critical information that our contributions claim.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Additional generalization evidence for our third contribution: performance on long-form video understanding benchmarks with Qwen2.5-VL-7B for 64 and 128 input frames (%), showing that the advantage extends to a third base model.

To further stress-test the generalization claimed in our third contribution, we ran additional experiments on the recently released Qwen2.5-VL model, evaluating on VideoMME and LongVideoBench to check that our method’s efficacy is not tied to a single architecture.

The results in Table 4 show that ST-SimDiff again performs strongly. At a 30% token retention ratio it scores 62.4 on VideoMME and 56.8 on LongVideoBench, and at a 50% ratio it improves to 62.9 and 58.4, respectively. Notably, at 50% the VideoMME score matches the upper bound obtained with all tokens, and the LongVideoBench score is highly comparable to its full-performance counterpart. These findings reinforce the broad generalization ability of our framework across base models.

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Ablation of the two halves of our second contribution on the NVILA-Video-8B model (%), reproducing on a second base model the pattern that both similarity and difference selection contribute.

C ABLATION STUDY RESULTS ON NVILA

To confirm that both halves of our second contribution generalize across base models, we repeat the component ablation on the NVILA-Video-8B model (Liu et al., 2024c). The setup mirrors the main-paper ablation: we start from a Baseline that uses only fundamental importance-based pruning and progressively add the similarity selection module (+ Sim) and the difference selection module (++ Diff).

As shown in Table 5, the results again validate our framework design, with trends highly consistent with LLaVA-Video-7B. Adding the similarity module (+ Sim) on top of the baseline steadily improves performance at both 30% and 50% token retain ratios. Comparing its three variants, spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.), the joint strategy is the most effective: at $r = 50\%$ it reaches 61.4% on VideoMME, ahead of the spatial-only and temporal-only variants, reaffirming that collaboratively modeling spatio-temporal associations is key to identifying redundant information. Adding the difference selection module (++ Diff) on top of the best similarity configuration completes our ST-SimDiff framework and, as the table shows, delivers the largest single gain. This provides cross-model evidence for both core modules of our contribution, similarity and difference selection, and once again validates our central idea of handling redundancy and change at the same time.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

The similarity threshold τ_{sim} and the difference threshold τ_{diff} are the two hyperparameters that govern, respectively, the representative and transitional token selection at the heart of our contribution, so their robustness matters. Figure 6 reports an ablation over both. Each follows the same rise-then-fall trend while remaining robust across a broad range: for τ_{sim} , too low a value yields imprecise community detection and too high a value fragments semantic clusters, while for τ_{diff} , too low a value admits noise through over-sensitivity and too high a value causes the model to miss key events. Performance is optimal at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, and we adopt this configuration throughout the rest of our experiments.

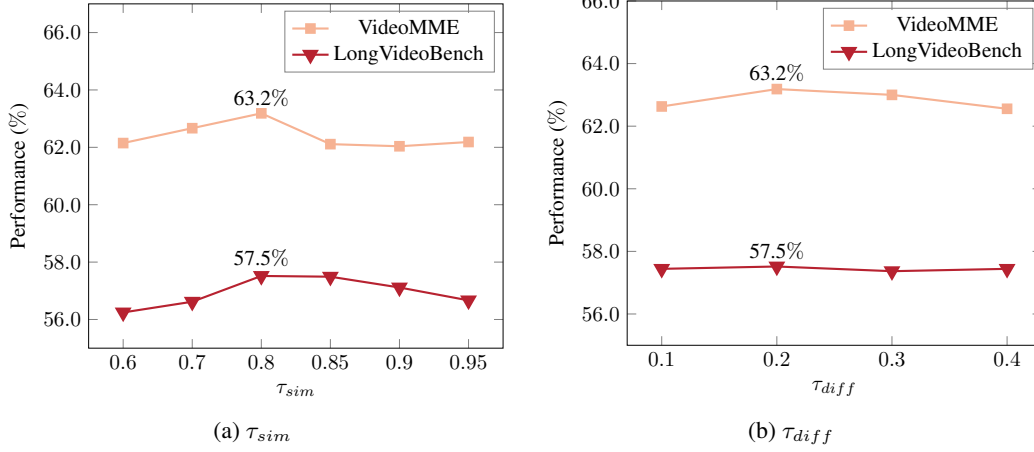


Figure 5: Robustness of the two thresholds central to our contribution: ablation over τ_{diff} and τ_{sim} on VideoMME and LongVideoBench, showing stable performance across a wide range.

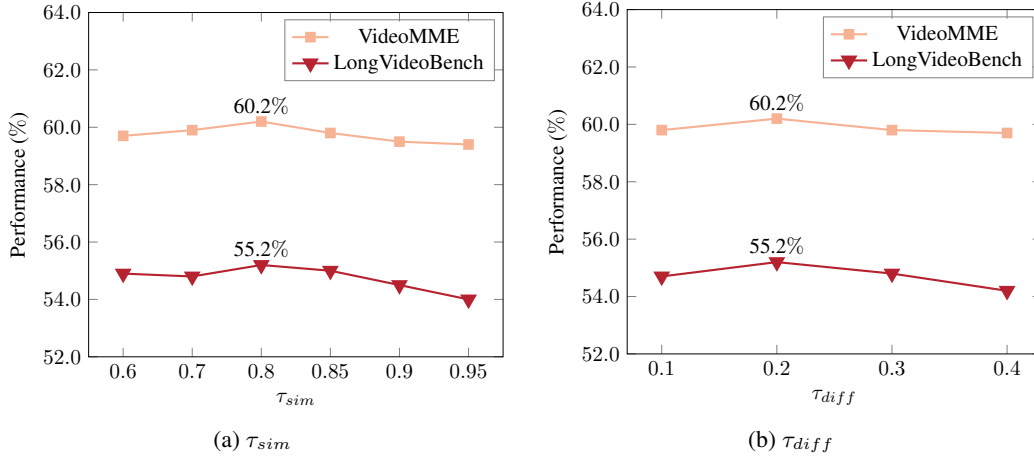


Figure 6: Threshold robustness of our contribution on a second base model: ablation over τ_{diff} and τ_{sim} on NVILA-Video-8B, with the same stable trend as on LLaVA-Video.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

To show that the threshold robustness underpinning our contribution holds beyond a single model, we repeat the hyperparameter ablation on NVILA-Video-8B. As shown in Fig. 6, we sweep the similarity threshold τ_{sim} and the difference threshold τ_{diff} on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . Sweeping τ_{sim} over $[0.6, 0.95]$, we find performance largely insensitive to this threshold. Peak performance is reached at $\tau_{sim} = 0.8$ on both benchmarks, 60.2% on VideoMME and 55.2% on LongVideoBench. Performance dips only slightly at the extremes, so the overall fluctuation is minimal, confirming that our method stays stable across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . Sweeping τ_{diff} over $[0.1, 0.4]$ shows the same strong robustness. Optimal performance again occurs around $\tau_{diff} = 0.2$, with VideoMME at 60.2% and LongVideoBench at 55.2%, indicating that a difference threshold near 0.2 best captures key events while compressing tokens.

In summary, this ablation confirms that the key hyperparameters of our contribution are robust and justifies our choice of $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the main experiments, as these values consistently give optimal performance across benchmarks and base models.